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# Dynamic Analysis of Planar Mechanisms With Fuzzy Joint Clearance and Random Geometry

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*The dynamic characteristics of planar mechanisms with fuzzy joint clearance and random geometry are studied in this paper. The dynamics model for the mechanism is constructed by utilizing Baumgarte approach in which the clearance size is a fuzzy number, while the geometry parameters are assumed as random variables. A hybrid contact force model, which consists of the Lankarani–Nikravesh model, improved Winkler elastic foundation model and modified Coulomb friction force model, is applied to construct revolute clearance joint. In order to solve the dynamics model, two methodologies are developed: confidence region method for quantification of random and fuzzy uncertainties (CRMQRFU) and confidence region method with transformation method (CRMTM). In the CRMQRFU, fuzzy numbers are first decomposed into intervals under the given membership level. Then, a general framework is proposed for quantification of random and interval uncertainties in the mechanism. In the CRMTM, a transformation method is applied to transform intervals into deterministic arrays, while probability theory is used to obtain the confidence regions under the given fuzzy values. The confidence region, considering random and fuzzy uncertainties, is obtained by fuzzy set theory. Finally, two examples are used to demonstrate the validity and feasibility of these methods.*
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## 1 Introduction

Uncertainty and clearance are unavoidable in mechanisms, due to abrasion, model inaccuracy, measurement error, assembly error, or manufacturing error, which result in deviation in predicting dynamic responses of the mechanisms. An accurate evaluation of the effects of uncertainty and clearance on the dynamic characteristics of mechanisms is an urgent problem in areas of mechanism design.

Rao and Bhatti [1] developed a probabilistic model to investigate the effects of manipulator parameters, following Gaussian distribution, on the kinematics and dynamics performance of a robotic manipulator. Batou and Soize [2] developed a random model of rigid bodies, accounting uncertainty of geometrical properties, for multibody dynamics, in which the prior probability density function of the stochastic model is obtained by the Maximum entropy principle. Wang et al. [3] investigated the dynamic characteristics of a flexible mechanism with random parameters in which Monte Carlo method is applied to calculate the mean and variance of the dynamic responses. Sandu et al. [4] applied the generalized polynomial chaos theory to construct mechanisms with uncertainty. Wu et al. [5] analyzed the dynamic behaviors of a rigid-flexible slider-crank mechanism in which the rigid and flexible components were represented as random variable and random field, respectively. Luo and Du [6] developed a probability method for probabilistic analysis of mechanisms with truncated random variables. Since joint clearance can significantly affect the kinematics and dynamic characteristics of mechanisms [7,8], Yan and Guo [9] investigated the kinematic accuracy of a flexible four-bar mechanism with uncertain clearances and link lengths. With the considerations of joint clearance and parameter uncertainty, Li et al. [10] investigated the kinematic accuracy and dynamic behaviors of a space deployable mechanism.

In some cases, sufficient data may not be easily available for defining a probability distribution due to the scarce of the information, which leads to the wide existence of epistemic uncertainty in mechanisms. Wu and Rao [11] evaluated the position accuracy of a Stanford manipulator by using interval method. In order to reduce wrapping effect and obtain tighter bounds of the solutions, Wu et al. [12] developed a Chebyshev inclusion functions-based interval method for dynamic analysis of rigid mechanisms with interval uncertainty. Guerine et al. [13] investigated the dynamic behaviors of a wind turbine gear system with interval parameters. They found that the response region obtained by the probabilistic approach is contained in that produced by the interval method. Wang et al. [14] analyzed the dynamic behaviors of a rigid-flexible multibody system with uncertain interval parameters by using Chebyshev surrogate model. Sun and Chen [15] developed a confidence region method for kinematic accuracy analysis of a slider-crank mechanism with random and interval uncertainties in which improved double-loop Monte Carlo sampling approach is applied for uncertainty propagation. Wang et al. [16] investigated the dynamic behaviors of flexible multibody systems with random and interval uncertainties.

In the previous studies, interval is used to represent epistemic uncertainty. In many cases, a designer may want to use "about" to represent an approximate parameter. For example, a designer may represent a dimension of "about 20 mm." For the imprecise descriptions, fuzzy sets are always used to represent the uncertainty. Till now, few researches are focused on the study of dynamic analysis of mechanisms with fuzzy uncertainty. Wasfy and Noor [17] used vertex method to predict the dynamic responses of a flexible four-bar linkage with fuzzy uncertainty. Wu and Rao [18] applied interval method for the fuzzy analysis of a four-bar mechanism with tolerances and clearances. More generally, random and fuzzy uncertainties exist in mechanisms

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simultaneously. To the best of our knowledge, there is no method which can analyze the dynamic behaviors of this kind of mechanisms. For a mechanism with random and fuzzy uncertainties, there exist two challenges when dealing with them [19]. First, an appropriate mathematical model should be selected to represent each of these uncertainties. Second, the influences of random and fuzzy uncertainties on the results should be analyzed separately. To solve the problem, we studied the dynamic behaviors of mechanisms with fuzzy joint clearance and random geometry, and presented two general methods for dynamic analysis of mechanisms with random and fuzzy uncertainties. Finally, the validity and the feasibility of these methods are demonstrated.

## 2 Equations of Motion for Planar Mechanism With Revolute Clearance Joint and Uncertainty

Using the Lagrange multipliers technique, the motion equations for a planar mechanism with revolute clearance joint, in which the constraint equation is added, can be expressed as

$$\begin{bmatrix} \mathbf{M} & \boldsymbol{\Phi}_{\mathbf{q}}^{T} \\ \boldsymbol{\Phi}_{\mathbf{q}} & 0 \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}^{\mathrm{A}} + \mathbf{F}_{\mathrm{C}} \\ \boldsymbol{\gamma} \end{bmatrix} \tag{1}$$

where $\ddot{\mathbf{q}}$ denotes the generalized acceleration vector. $\mathbf{M}$ is the generalized mass matrix. $\boldsymbol{\lambda}$ is the Lagrange multiplier vector. $\boldsymbol{\Phi}_{\mathbf{q}}$ is the constraint Jacobian matrix. $\boldsymbol{\gamma} = -(\boldsymbol{\Phi}_{\mathbf{q}}\dot{\mathbf{q}})_{\mathbf{q}} - 2\boldsymbol{\Phi}_{\mathbf{q}t}\dot{\mathbf{q}} - \boldsymbol{\Phi}_{tt}$. $\mathbf{Q}^{\mathrm{A}}$ is the external force. $\mathbf{F}_{\mathrm{C}}$ is the contact force.

In mechanism design, uncertainty is composed of random uncertainty and imprecision [20,21]. Random uncertainty derives from an inherent random nature of physical quantities, which is irreducible and usually represented with probability distribution. Imprecision is used to mean uncertainty in choosing among alternatives, which is usually represented using fuzzy sets [22–24]. More generally, deterministic, random, and fuzzy variables exist in the mechanism simultaneously. Furthermore, to avoid the constraints violation in Eq. (1), the Baumgarte approach [25] is applied. Thus, Eq. (1) is modified as

$$\begin{bmatrix} \tilde{\mathbf{M}} & \tilde{\boldsymbol{\Phi}}_{\mathbf{q}}^{T} \\ \tilde{\boldsymbol{\Phi}}_{\mathbf{q}} & 0 \end{bmatrix} \begin{bmatrix} \ddot{\tilde{\mathbf{q}}} \\ \tilde{\boldsymbol{\lambda}} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}^{\mathrm{A}} + \tilde{\mathbf{F}}_{\mathrm{C}} \\ \tilde{\boldsymbol{\gamma}} - 2\bar{\alpha}\dot{\tilde{\boldsymbol{\Phi}}} - \beta^{2}\tilde{\boldsymbol{\Phi}} \end{bmatrix} \tag{2}$$

where $\bar{\alpha}$ and $\beta$ are the Baumgarte parameters. The overtilde "$\sim$" expresses the random and fuzzy variables.

## 3 Contact Force in Revolute Clearance Joint

**3.1 Modeling of Revolute Clearance Joint.** In this section, a mathematical model is constructed to simulate the revolute clearance joint, as shown in Fig. 1, which consists of a bushing fixed to the body $i$ and a pin fixed to the body $j$. $x_i o_i y_i$ and $x_j o_j y_j$ are the local coordinate systems attached to the center of each body, respectively. Points $P_i$ and $P_j$ are the centers of the bearing and journal, respectively. $\mathbf{e}_{ij}$ is the eccentricity vector connecting the centers of the bearing and journal, which is given as

$$\mathbf{e}_{ij} = \mathbf{r}_j^{p} - \mathbf{r}_i^{p} = (\mathbf{r}_j + \mathbf{A}_j s_j') - (\mathbf{r}_i + \mathbf{A}_i s_i') \tag{3}$$

where $\mathbf{r}_i^{p}$ and $\mathbf{r}_j^{p}$ are the position vectors of $P_i$ and $P_j$ defined in the global coordinates. $s_i'$ and $s_j'$ are local position vectors that link the centers of masses to the centers of the bushing and pin in the local coordinate system. $\mathbf{A}_i$ and $\mathbf{A}_j$ are the rotational transformation matrices.

The unit eccentricity vector is expressed as

$$\mathbf{n} = \frac{\mathbf{e}_{ij}}{e_{ij}} \tag{4}$$

where $e_{ij}$ is the magnitude of the eccentricity vector. The relative penetration depth ($\delta$) is given as

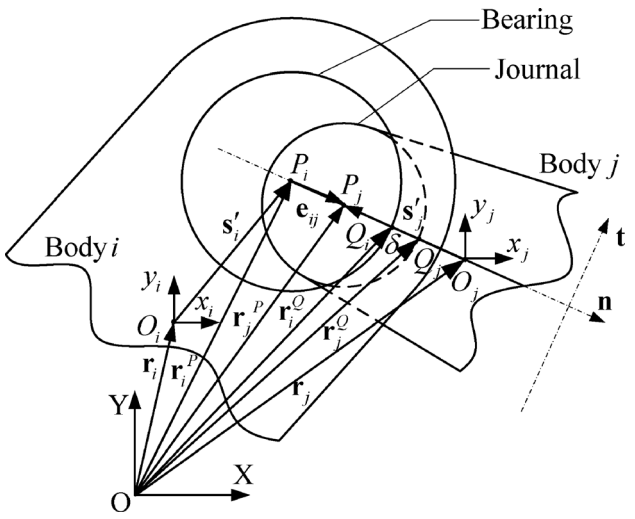



**Fig. 1 Schematic illustration of revolute clearance joint**

$$\delta = e_{ij} - c \tag{5}$$

where $c$ is the radial clearance size, $c = R_i - R_j$. $R_i$ and $R_j$ are the radii of the bearing and journal, respectively. As shown in Fig. 1, points $Q_i$ and $Q_j$ represent the contact points on body $i$ and $j$, respectively. The position of the two points can be given as

$$\mathbf{r}_k^{Q} = \mathbf{r}_k + \mathbf{A}_k s_k' + R_k \mathbf{n} (k = i, j) \tag{6}$$

The velocity of the contact points in the global coordinate system can be obtained by differentiating Eq. (6) with respect to time, yielding

$$\dot{\mathbf{r}}_k^{Q} = \dot{\mathbf{r}}_k + \dot{\mathbf{A}}_k s_k' + R_k \dot{\mathbf{n}} (k = i, j) \tag{7}$$

The relative scalar velocities, normal and tangential to the plane of collision, are given by

$$v_{\mathrm{N}} = (\dot{\mathbf{r}}_i^{Q} - \dot{\mathbf{r}}_j^{Q})^{\mathrm{T}} \mathbf{n} \tag{8}$$

$$v_{\mathrm{T}} = (\dot{\mathbf{r}}_i^{Q} - \dot{\mathbf{r}}_j^{Q}) \mathbf{t} \tag{9}$$

where the tangent vector $\mathbf{t}$ can be obtained by rotating vector $\mathbf{n}$ by 90 deg counter-clockwise.

**3.2 Contact Force Model.** Contact force occurs with the contact between bearing and journal. Hertz model is the most well-known contact force model, which is restricted to perfectly elastic solids. Based on the Hertz model, Lankarani and Nikravesh, Herbert and McWhannell, and Lee et al. developed various contact force models in which the energy dissipated during the impact is represented [26]. Liu et al. [27] point out that Hertz model is available only in the case of large joint clearance and small normal load. But, in engineering applications, clearance in actual revolute joint is always very small, which certainly affects the accuracy of the results. On the other hand, most researchers take the contact stiffness and damping coefficient as constants for the sake of simplicity, which is not according to actual situations. To solve these problems, Bai and Zhao [28] developed a hybrid contact force model consisting of the Lankarani–Nikravesh model and improved Winkler elastic foundation model. The contact force model has the form of

$$F_{\mathrm{N}} = K_n \delta^{n} + D_{\mathrm{mod}} \dot{\delta} \tag{10}$$

where $F_{\mathrm{N}}$ is the normal contact force and $\dot{\delta}$ denotes the relative penetration velocity. The nonlinear stiffness coefficient ($K_n$) and the modified damping coefficient ($D_{\mathrm{mod}}$) are expressed as



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$$K_n = \frac{1}{8}\pi E^* \sqrt{\frac{2\delta\left(3R_i - 3R_j + 2\delta\right)^2}{\left(R_i - R_j + \delta\right)^3}} \tag{11}$$

$$D_{\text{mod}} = \frac{3K_n\left(1 - c_e^2\right)e^{2(1-c_e)}\delta^n}{4\dot{\delta}^{(-)}} \tag{12}$$

where $\dot{\delta}^{(-)}$ denotes the initial penetration velocity. $c_e$ is the restitution coefficient. $E^*$ is the compound elastic modulus and is represented as

$$\frac{1}{E^*} = \frac{1 - v_i^2}{E_i} + \frac{1 - v_j^2}{E_j} \tag{13}$$

where $E_k(k = i, j)$ and $v_k(k = i, j)$ are Young's modulus and Poisson's coefficient, respectively. In the hybrid contact force model, nonlinear stiffness and modified damping coefficient vary with penetration depth during the contact process which is more realistic [28]. Thus, the hybrid contact force model is more precise than the Hertz model in which contact stiffness is taken as a constant.

**3.3 Friction Force Model.** The most well-known model, Coulomb's friction model, is considered to be reliable in many instances in the modeling of mechanisms. In Coulomb law, the friction force is assumed to be proportional to the normal load and has an opposite direction to the relative motion between the two contacting bodies. The friction force depends on the direction of the relative velocity. It is worth noting that the main disadvantage when using this model for simulation is to determine when the velocity is zero, and the exact value $v = 0$ is never exactly matched during numerical integration. Moreover, some care must be used to deal with the discontinuities during numerical simulation. In order to keep the continuity of the friction force–velocity relation at zero relative velocity point, Ambrósio [29] proposed a modified Coulomb's friction law. The tangential friction force ($F_T$), shown in Fig. 2, is given as

$$F_T = -c_f c_d F_N \frac{\mathbf{v}_T}{v_T} \tag{14}$$

where $c_f$ is the friction coefficient and $\mathbf{v}_T$ denotes the relative tangential velocity. The dynamic correction coefficient $c_d$ is given as

$$c_d = \begin{cases} 0 & \text{if} \quad v_T \le v_0 \\ \dfrac{v_T - v_0}{v_1 - v_0} & \text{if} \quad v_0 \le v_T \le v_1 \\ 1 & \text{if} \quad v_T \ge v_1 \end{cases} \tag{15}$$

where $v_0$ and $v_1$ are the given tolerances for the velocity.

## 4 Methodologies for Quantification of Random and Fuzzy Uncertainties

Equation (2) is a differential-algebraic equation with random and fuzzy uncertainties, which can only be solved by using numerical method. To the best of our knowledge, there is no current uncertainty analysis method which can be directly used to solve this kind of dynamics equation. In this section, two new methods will be proposed to solve this kind of dynamics equation: confidence region method for quantification of random and fuzzy uncertainties (CRMQRFU) and confidence region method with transformation method (CRMTM). Previously, the concept of the fuzzy number and the α-cut approach are first introduced.

**4.1 Fuzzy Sets and Numbers.** The concept of a fuzzy set as introduced by Zadeh [30] avoids the problem of classical set theory with the element either belonging to a set (the membership level is 1) or not (the membership level is 0). A membership level $\mu_{\tilde{A}}$ of an element $x$ in a fuzzy set $\tilde{A}$ can range from 0 to 1, which is assigned depending on the designer's desires or engineering data [31]. A fuzzy set $\tilde{A}$ can be defined as

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) | x \in X, \mu_{\tilde{A}}(x) \in [0, 1]\} \tag{16}$$

A fuzzy number is a normal and convex fuzzy set with continuous membership function. And there exists only one $x$ yielding to $\mu_{\tilde{A}}(x) = 1$ [30]. Figure 3 expresses a triangular fuzzy number, which is defined by its membership function. The α-cuts of $\tilde{A}$ are always used to represent the fuzzy number and the membership function is subdivided into α-cuts at the α-levels $\mu_j$ as

$$\mu_j = j \cdot \Delta\mu, \quad \text{for } j = 0, 1, \ldots, m, \quad \text{where } \Delta\mu = 1/m \tag{17}$$

At each level $\mu_j = \alpha$, the confidence interval is given as $[a_j, b_j]$ in which the membership value $\mu_j \ge \alpha$. The α-cut for a triangular fuzzy number is shown in Fig. 3. Then, the all $m + 1$ intervals are applied to describe the fuzzy number. Thus, the issue of dynamic analysis for a mechanism with fuzzy uncertainty is transformed to that with interval uncertainty.

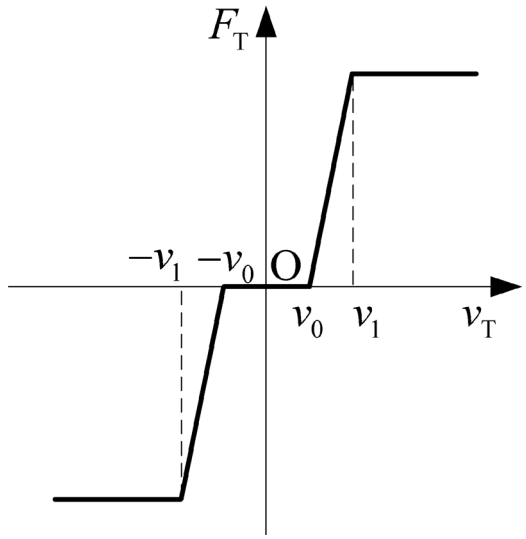



**Fig. 2 The modified Coulomb friction force model**

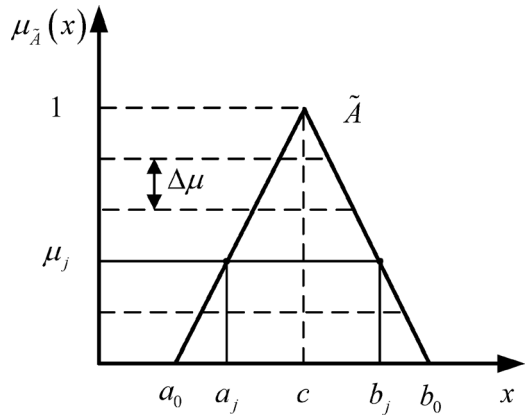



**Fig. 3 The α-cut for a triangular fuzzy number with membership function $\mu\tilde{A}$ at membership level**



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**4.2 Confidence Region Method for Quantification of Random and Fuzzy Uncertainties.** The dynamic response of a mechanism with random and fuzzy uncertainties is expressed as

$$y(t) = f(t, \mathbf{a}_N, \mathbf{e}_M) \tag{18}$$

where $t$ is the time. $\mathbf{a}_N = [a_1, \ldots, a_{nA}]$ includes all the $nA$ random variables. $\mathbf{e}_M = [e_{M1}, \ldots, e_{M,nM}]$ includes all the $nM$ fuzzy numbers. Since fuzzy numbers can be transformed to intervals, according to $\alpha$-cuts approach, the dynamic response of the mechanism can be rewritten as

$$\mathrm{y}^{\alpha}(t) = f(t, \mathbf{a}_N, \mathbf{e}_M^{\alpha}) \tag{19}$$

where $\mathbf{e}_M^{\alpha}$ is the $\alpha$-cut for the fuzzy numbers. $f(t, \mathbf{a}_N, \mathbf{e}_M^{\alpha})$ only contains random variables and interval variables.

With $\mathbf{e}_M^{\alpha}$ fixed as indicated, the cumulative distribution function (CDF) for $y^{\alpha}(t)$ at a particular moment $t_j$ is defined by plots of the points $[y^{\alpha}(t_j), \mathrm{Pr}ob_A(\tilde{y} \le y^{\alpha}(t_j), \mathbf{e}_M^{\alpha})]$ with

$$\begin{aligned} \mathrm{Pr}\,ob_A(\tilde{y} \le y^{\alpha}(t_j), \mathbf{e}_M^{\alpha}) &= \int_{\Lambda} \underline{\delta}_y[f(t_j, \mathbf{a}, \mathbf{e}_M^{\alpha})] d_A(\mathbf{a}) dA \\ &\cong \sum_{i=1}^{\mathrm{nSA}} \underline{\delta}_y[f(t_j, \mathbf{a}_i, \mathbf{e}_M^{\alpha})]/\mathrm{nSA} \end{aligned} \tag{20}$$

where $\mathbf{a}_i, i = 1, 2, \cdots, \mathrm{nSA}$, is a random sample generated from the possible values for $\mathbf{a}$ in consistency with the density function $d_A(\mathbf{a})$. $\underline{\delta}_y$ is defined as

$$\underline{\delta}_y[f(t_j, \mathbf{a}, \mathbf{e}_M^{\alpha})] = \begin{cases} 1 & \text{if } f(t_j, \mathbf{a}, \mathbf{e}_M^{\alpha}) \le y \\ 0 & \text{otherwise} \end{cases} \tag{21}$$

Different values for $\mathbf{e}_M^{\alpha}$ result in different CDFs of $y^{\alpha}(t_j)$. In order to express these CDFs, an uncertainty structure is defined as

$$\underline{\mathrm{P}}(y^{\alpha}(t_j)) = \{p : p = \mathrm{Pr}ob_A(\tilde{y} \le y^{\alpha}(t_j), \mathbf{e}_M^{\alpha}) \text{ for } \mathbf{e}_M^{\alpha} \in \boldsymbol{\Xi}\} \tag{22}$$

$\underline{\mathrm{P}}(y^{\alpha}(t_j))$ is the set of all probabilities for a value $\tilde{y}$ not greater than $y$, which is derived from the CDFs as indicated in Eq. (20). $\boldsymbol{\Xi}$ contains all possible values for $\mathbf{e}_M^{\alpha}$.

The value $\mathrm{Pr}b_q(y^{\alpha}(t_j))$ for the $q$ quantile of the set $\underline{\mathrm{P}}(y^{\alpha}(t_j))$ is calculated by [19]

$$\mathrm{Pr}b_q(y^{\alpha}(t_j)) = \inf\{p : p \in \underline{\mathrm{P}}(y^{\alpha}(t_j)) \text{ and } \mathrm{Pr}ob_E[\underline{\mathrm{P}}_p(y^{\alpha}(t_j))] \ge q\} \tag{23}$$

where $\mathrm{Pr}ob_E[\underline{\mathrm{P}}_p(y^{\alpha}(t_j))]$ is distribution for each set $\underline{\mathrm{P}}_p(y^{\alpha}(t_j))$. $\underline{\mathrm{P}}_p(y^{\alpha}(t_j))$ is expressed as

$$\underline{\mathrm{P}}_p(y^{\alpha}(t_j)) = \{\tilde{p} : \mathbf{e}_M^{\alpha} \in \boldsymbol{\Xi} \text{ and } \tilde{p} = \mathrm{Pr}ob_A(\tilde{y} \le y^{\alpha}(t_j), \mathbf{e}_M^{\alpha}) > p\} \tag{24}$$

$\underline{\mathrm{P}}_p(y^{\alpha}(t_j))$ defines a set that $\mathrm{Pr}ob_A(\tilde{y} \le y^{\alpha}(t_j), \mathbf{e}_M^{\alpha})$ is larger than $p$. It is a nonincreasing function of $p$ because $\underline{\mathrm{P}}_v(y^{\alpha}(t_j)) \subseteq \underline{\mathrm{P}}_u(y^{\alpha}(t_j))$ for $0 \le u \le v \le 1$.

The confidence region of $y^{\alpha}(t)$ with 95% confidence level is defined by

$$\begin{aligned} \Psi(\mathrm{y}^{\alpha}(t)) = \{[y(t)_d y(t)_u] : \mathrm{Pr}b_0(y^{\alpha}(t)_d) = 2.5\% \text{ and} \\ \mathrm{Pr}b_1(y^{\alpha}(t)_u) = 97.5\%, t \in \mathrm{T}\} \end{aligned} \tag{25}$$

where $T$ is the time interval of interest. $y(t)_u$ and $y(t)_d$ are the upper and lower bounds of the 95% confidence region, respectively.

The 95% confidence region of $\mathrm{y}^{\alpha}(t)$ at different $\alpha$-levels can be easily obtained by Eq. (25) with various $\alpha$-cut for the fuzzy numbers.

**4.3 Confidence Region Method With Transformation Method.** Transformation method [32–34] has been widely used to evaluate dynamic response of mechanical systems for its powerful and practical implementation. However, traditional transformation method can only be used to analyze the dynamic behaviors of mechanisms with fuzzy uncertainty. For mechanisms with random and fuzzy uncertainties, this method cannot be used directly. Considering that, a novel method for dynamic analysis of mechanisms with random and fuzzy uncertainties is proposed. In the method, each fuzzy number $e_{Mi}, i = 1, 2, \ldots, nM$, is first decomposed into $m+1$ intervals $X_i^{(j)}, j = 0, 1, \ldots, m$, leading to a set

$$P_i = \{X_i^{(0)}, X_i^{(1)}, \ldots, X_i^{(m)}\} \tag{26}$$

with

$$X_i^{(j)} = [a_i^{(j)}, b_i^{(j)}], \;\; a_i^{(j)} \le b_i^{(j)}, \;\; i = 1, 2, \ldots, nM, \;\; j = 0, 1, \ldots, m$$

where $a_i^{(j)}$ and $b_i^{(j)}$ are the lower and upper bounds of the interval at the membership level $\mu^j$ for the $i$th fuzzy number, respectively.

Then, the intervals can be transformed into arrays $\hat{X}_i^{(j)}$ as

$$\hat{X}_i^{(j)} = \underbrace{(\alpha_i^{(j)}, \beta_i^{(j)}, \alpha_i^{(j)}, \beta_i^{(j)}, \ldots, \alpha_i^{(j)}, \beta_i^{(j)})}_{2^{i-1}\text{pairs}} \tag{27}$$

with

$$\alpha_i^{(j)} = \underbrace{(a_i^{(j)}, a_i^{(j)}, \ldots, a_i^{(j)})}_{2^{nM-i}\text{elements}}, \quad \beta_i^{(j)} = \underbrace{(b_i^{(j)}, b_i^{(j)}, \ldots, b_i^{(j)})}_{2^{nM-i}\text{elements}}$$

For membership level $\mu^j, j = 0, 1, \ldots, m-1$, there are $2^{nM}$ combinations, and there is only one combination $(c_1, c_2, \ldots, c_{nM})$ need to be evaluated. Thus, $1 + m \cdot 2^{nM}$ combinations, for all membership levels and fuzzy numbers, need to be evaluated by probability theory. The output $f(t, \mathbf{a}_N, \mathbf{e}_M)$ of the system is then transformed to the region arrays $\hat{\mathrm{Z}}_r^{(j)} = ({}^1\hat{z}_r^{(j)}, {}^2\hat{z}_r^{(j)}, \ldots, {}^{2^{nM}}\hat{z}_r^{(j)}), j = 0, 1, \ldots, m$. The $k$th element of the arrays $\hat{\mathrm{Z}}_r^{(j)}$ can be expressed as

$${}^k\hat{z}_r^{(j)} = f_r(t, \mathbf{a}_N, {}^k\hat{\mathrm{x}}_1^{(j)}, {}^k\hat{\mathrm{x}}_2^{(j)}, \ldots, {}^k\hat{\mathrm{x}}_n^{(j)}) \tag{28}$$

where ${}^k\hat{\mathrm{x}}_i^{(j)}$ is the $kth$ element of the arrays $\hat{X}_i^{(j)}$. Unlike traditional transformation method, in which ${}^k\hat{z}_r^{(j)}$ is scalar, in this method, ${}^k\hat{\mathrm{z}}_r^{(j)}$ is the confidence region of the dynamic response under the given fuzzy variables, obtained by the probabilistic approach.

The region arrays $\hat{\mathrm{Z}}_r^{(j)}$ are retransformed to the output confidence region $\mathrm{Z}_r^{(j)}$ for each membership level $\mu_j$, where

$$\mathrm{Q}_r = \{\mathrm{Z}_r^{(0)}, \mathrm{Z}_r^{(1)}, \ldots, \mathrm{Z}_r^{(m)}\} \tag{29}$$

The output confidence region $\mathrm{Z}_r^{(j)}$ is obtained by retransforming the arrays $\hat{\mathrm{Z}}_r^{(j)}$ via the recursive procedure

$$\mathrm{Z}_r^{(j)} = \cup(\mathrm{Z}_r^{(j+1)}, \hat{\mathrm{Z}}_r^{(j)}), \quad j = 0, 1, \ldots, m-1 \tag{30}$$

and

$$\mathrm{Z}_r^{(m)} = \hat{\mathrm{Z}}_r^{(m)} \tag{31}$$

where $U$ denotes union of confidence regions. It means that the confidence region at the core level, considering random and fuzzy uncertainties, is equal to the union of the confidence regions at the core level, considering random uncertainty, and the confidence region at a higher $\alpha$-sublevel.



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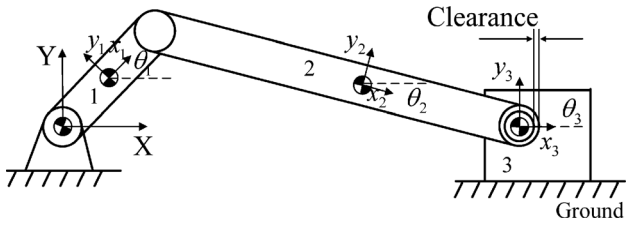



**Fig. 4 Slider-crank mechanism with clearance**

**Table 1 Parameters used in the dynamic simulation of the slider-crank mechanism**

| | | | |
|---|---|---|---|
| Bearing radius | 10.0 mm | Baumgarte-$\alpha$, $\beta$ | 250 |
| Restitution coefficient | 0.9 | Friction coefficient | 0.05 |
| Young's modulus | 207 GPa | Integration step size | 0.00001 s |
| Poisson's ratio | 0.3 | Integration tolerance | 0.000001 |

## 5 Results and Discussion

**5.1 Slider-Crank Mechanism With Clearance.** A planar slider-crank mechanism with revolute clearance joint is chosen to demonstrate the application of the proposed methods. The schematic illustration of the mechanism, consisting of three rigid bodies (crank 1, connecting rod 2, and slider 3), is shown in Fig. 4. The global generalized coordinates of the three bodies are $\mathbf{q} = [x_1, y_1, \theta_1, x_2, y_2, \theta_2, x_3, y_3, \theta_3]^{\mathrm{T}}$. Revolute or translational joints are applied at the connections. Imperfect joints are assumed to be located at the connection of the connecting rod and slider, while perfect joints are assumed at rest positions. The constraints corresponding to the slider-crank mechanism consist of seven equations expressed as

$$\mathbf{\Phi} = \begin{bmatrix} x_1 - 0.5l_1\cos\theta_1 \\ y_1 - 0.5l_1\sin\theta_1 \\ x_2 - l_1\cos\theta_1 - 0.5l_2\cos\theta_2 \\ y_2 - l_1\sin\theta_1 - 0.5l_2\sin\theta_2 \\ y_3 \\ \theta_3 \\ \theta_1 - \omega t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{32}$$

where $l_1$ and $l_2$ are the crank length and the connecting rod length, respectively. The first six constraints in Eq. (32) describe the perfect joints in the slider-crank mechanism. The final constraint imposes a constant angular velocity $\omega$ on the motion of the crank.

The mass and moment of inertia of the aluminum slider are assumed as 0.14 kg and $2 \times 10^{-6}$ kg m$^2$, respectively. The linear densities (mass per unit length) for the aluminum crank and connecting rod are 6 kg/m and 1.75 kg/m, respectively. The crank and connecting rod lengths are considered as uncertain parameters. The dynamic parameters used in simulations are listed in Table 1. The journal radius is considered as uncertain parameter and so is the clearance size. The crank rotates with a constant angular speed of 5000 rpm. The three uncertain parameters, clearance size, crank, and connecting rod lengths, are assumed to be independent

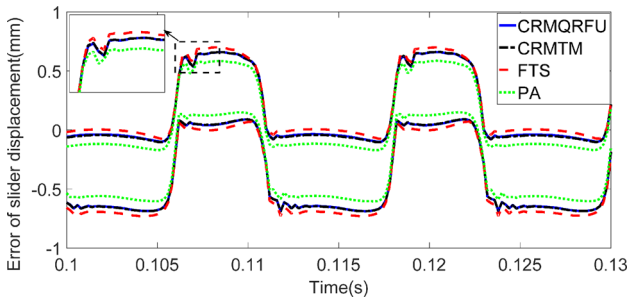



**Fig. 5 Slider displacement errors with different uncertain parameters at $\alpha$-cut 0**

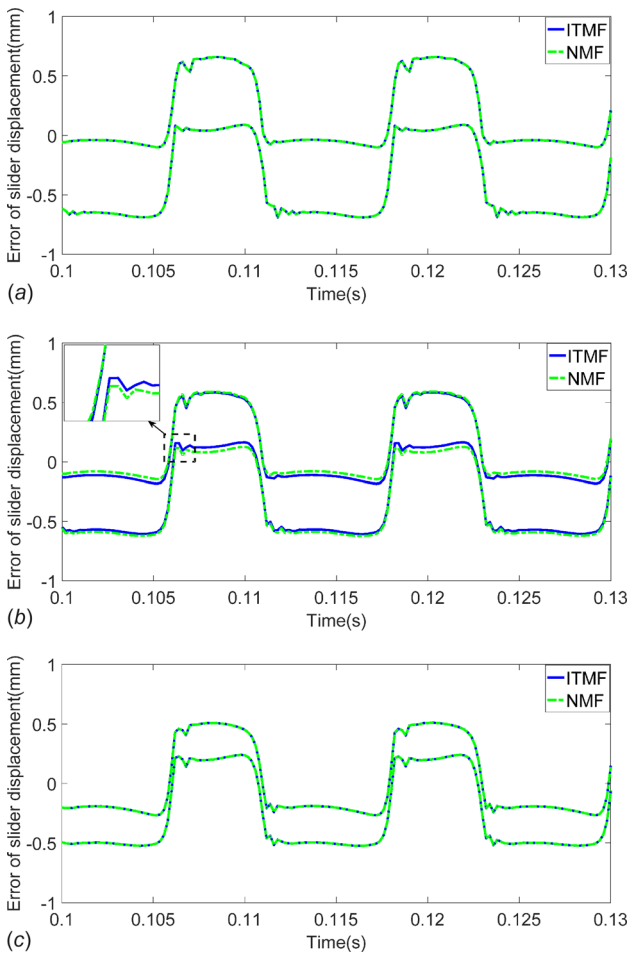



**Fig. 6 Slider displacement errors for isosceles triangular membership function and nonlinear membership function at cases of (*a*) $\alpha$-cut 0, (*b*) $\alpha$-cut 0.5, and (*c*) $\alpha$-cut 1, respectively**

with each other, and three uncertain situations are considered. In the first case, the crank and connecting rod lengths are assumed as random variables, following uniform distribution, while the clearance size is a fuzzy number, defined using an isosceles triangular membership function. In the second case, all the above three

**Table 2 Uncertain parameters of the slider-crank mechanism for different conditions**

| Parameter | Case 1 | Case 2 | Case 3 |
|---|---|---|---|
| Clearance size (mm) | {0.2, 0.35, 0.5} | $U$(0.2, 0.5) | {0.2, 0.35, 0.5} |
| Crank length (mm) | $U$(49.9, 50.1) | $U$(49.9, 50.1) | {49.9, 50, 50.1} |
| Connecting rod length (mm) | $U$(119.9, 120.1) | $U$(119.9, 120.1) | {119.9, 120, 120.1} |



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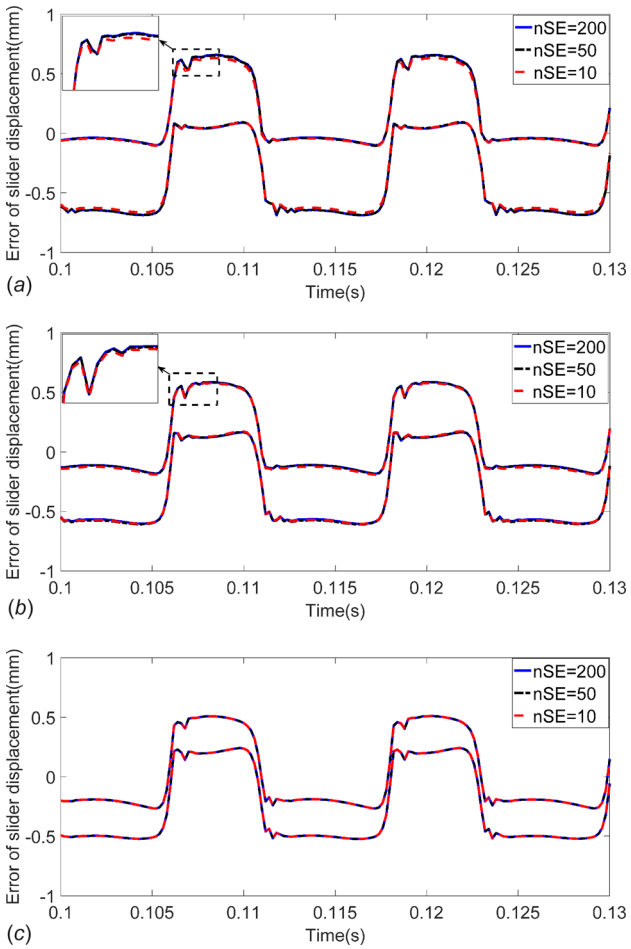



**Fig. 7 Slider displacement error for nSE = 10, 50, and 200 obtained by CRMQRFU at cases of (*a*) α-cut 0, (*b*) α-cut 0.5, and (*c*) α-cut 1, respectively**

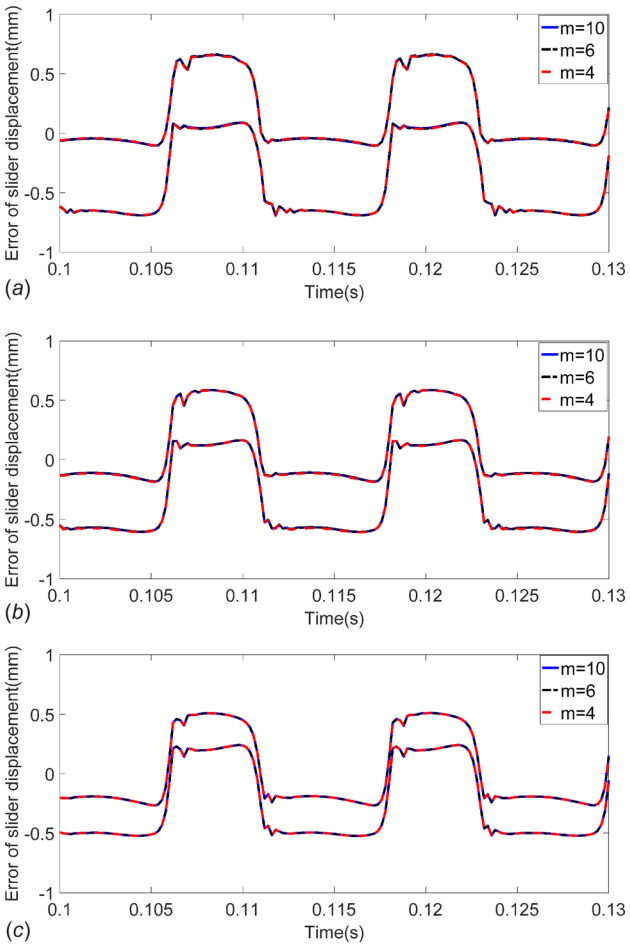



**Fig. 8 Slider displacement error for $m = 4$, 6, and 10 obtained by CRMTM at cases of (*a*) α-cut 0, (*b*) α-cut 0.5, and (*c*) α-cut 1, respectively**

parameters are uniform random variables. In the third case, the three parameters are fuzzy numbers with isosceles triangular membership functions. Table 2 shows the uncertain parameters of the slider-crank mechanism for different situations.

In the first case, the CRMQRFU and CRMTM are applied to obtain the 95% confidence region of the slider displacement error at α-cut 0. In the CRMQRFU, the improved double-loop Monte Carlo sampling approach [35] is used. The numbers used for sampling of the random and fuzzy uncertainties are 2000 and 200 (nSE), respectively. In the CRMTM, the fuzzy clearance size is subdivided into ten intermediate intervals ($m = 10$). The Monte Carlo sampling approach is used, and the number used for sampling of the random uncertainty is 2000. In the second case, the 95% confidence region of the slider displacement error is computed by the probabilistic approach. In the third one, the fuzzy set theory is used to obtain the response region of the slider displacement error at α-cut 0. The slider displacement errors obtained by the probabilistic approach, fuzzy set theory, CRMQRFU and CRMTM are shown in Fig. 5. It is seen that the confidence region obtained by the CRMQRFU tends to be close to that produced by the CRMTM. The confidence region for the second case is contained in the result for the first case. It indicates that the proposed methods yield larger bounds than the probabilistic approach. The confidence region for the first case is contained in that for the third case. It indicates that the proposed methods can obtain tighter bounds than the fuzzy set theory.

Now, we consider the case that the fuzzy clearance size has nonlinear membership function. The maximum and minimum values of the interval for a given α-cut are described for the clearance size as

$$c_{\alpha\min} = \frac{7}{20}\left(\frac{4}{7} + \frac{3}{7}\alpha^2\right), \quad c_{\alpha\max} = \frac{7}{20}\left(\frac{10}{7} - \frac{2}{7}\alpha - \frac{1}{7}\alpha^2\right) \tag{33}$$

The confidence regions of the slider displacement error with isosceles triangular membership function and nonlinear membership function at α-cut 0, 0.5, and 1, obtained by the CRMQRFU, are shown in Fig. 6. It can be observed that there is significant difference between the confidence regions with isosceles triangular membership function and nonlinear membership function at α-cut 0.5, while there is no difference between the confidence regions with isosceles triangular membership function and nonlinear membership function at α-cut 0 and 1. The explanation is that the intervals at the membership level 0 or 1 for isosceles triangular membership function and nonlinear membership function are the same, while there is difference at α-cut 0.5. Thus, the membership function of the clearance size can affect the results by α-cut.

To further investigate the effects of the nSE and $m$ on the results, in the CRMQRFU, nSE is selected as 10, 50, and 200. In the CRMTM, $m$ is selected as 4, 6, and 10. The confidence regions of the slider displacement errors at α-cut 0, 0.5, and 1, obtained by the CRMQRFU and CRMTM, are shown in Figs. 7 and 8, respectively. It can be seen that the results obtained by the CRMTM with $m = 4$, 6, and 10 tend to be close. The confidence region of the slider displacement error obtained by the CRMQRFU with $\mathrm{nSE} = 50$ tends to be close to that with $\mathrm{nSE} = 200$. However, there is a significant error between $\mathrm{nSE} = 50$ and $\mathrm{nSE} = 10$. It indicates that the confidence region of the slider displacement error for the mechanism obtained by the CRMTM with small number of α-levels can meet accuracy requirements. The results obtained by



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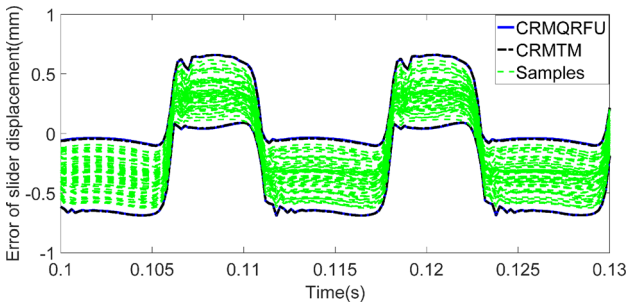



**Fig. 9 Slider displacement error obtained by CRMQRFU, CRMTM at $\alpha$-cut 0 and 50 random samples**

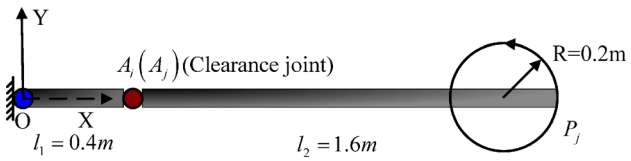



**Fig. 10 Circle trajectory tracking of the deployable mechanism with clearance joint**

**Table 3 Parameters for the dynamic analysis of the deployable mechanism**

| | | | |
|---|---|---|---|
| Journal radius | 10.0 mm | Baumgarte-$\alpha$, $\beta$ | 250 |
| Restitution coefficient | 0.9 | Friction coefficient | 0.1 |
| Young's modulus | 207 GPa | Integration step size | 0.00001 s |
| Poisson's ratio | 0.3 | Integration tolerance | 0.000001 |

the CRMQRFU can meet accuracy requirements in case that there are enough samples. The computational efficiency of the two methods are compared. The kinematic accuracy of the mechanism is simulated using MATLAB on a PC with an eight core central processing unit of 3.2 GHz and 32 G memories. The computing time of the CRMQRFU with nSE = 50 is 40352 s, while the computing time of the CRMTM with $m = 4$ is 11821 s. The calculated amount of the CRMTM is reduced to 29.3% of that for the CRMQRFU, which indicates a higher efficiency of CRMTM in calculation of this kind of problem.

To verify the reliability of the results, 50 random samples are simulated. As shown in Fig. 9, the slider displacement errors of the fifty samples are contained in the confidence regions of the slider displacement error, obtained by the CRMTM with $m = 10$ and CRMQRFU with nSE = 200, which validate the reliabilities of the results.

**5.2 Simple Planar Space Deployable Mechanism With Clearance.** The simple planar space deployable mechanism with clearance joint is controlled to track a circle trajectory, as shown in Fig. 10, with the designed angular velocity as follows:

$$\omega = -\frac{6\pi}{500}(t^2 - 10t) \tag{34}$$

The linear densities for the links $OA_i$ and $A_jP_j$ are 8 kg/m. In this work, the hybrid contact force model and the modified Coulomb friction force model are employed to construct the clearance joint. The parameters for the dynamic analysis of the deployable mechanism are shown in Table 3.

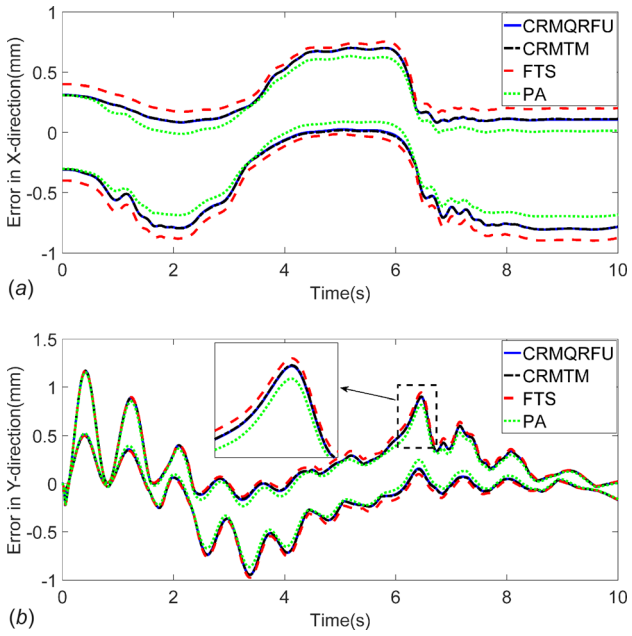



**Fig. 11 Trajectory errors of the mechanism for different uncertainty situations in (*a*) the *X*-direction and (*b*) the *Y*-direction, respectively**

The three uncertain parameters, clearance size, lengths of links $OA_i$ and $A_jP_j$, are assumed to be independent with each other. And three uncertain situations are investigated, separately. In the first case, the lengths for $OA_i$ and $A_jP_j$ are assumed as random variables, following uniform distribution. The clearance size is a fuzzy number, defined using an isosceles triangular membership function. In the second case, all the above three parameters are uniform random variables. In the third case, the three parameters are fuzzy numbers with isosceles triangular membership functions. Table 4 shows the uncertain parameters of the deployable mechanism for the three situations. The confidence regions of the trajectory errors of the mechanism with different uncertainty situations are given in Fig. 11. It is clear that the confidence regions obtained by the CRMQRFU and CRMTM are close. The confidence region obtained by the probabilistic approach is contained in the result obtained by the CRMQRFU, which indicates that the proposed methods yield larger bounds than the probabilistic approach. The confidence region obtained by the CRMQRFU is contained in that obtained by the fuzzy set theory, which indicates that the proposed methods can obtain tighter bounds than the fuzzy set theory.

## 6 Conclusions

Two general approaches for dynamic analysis of mechanisms with fuzzy joint clearance and random geometry have been developed. In the CRMQRFU, fuzzy numbers are decomposed into intervals, according to $\alpha$-cuts approach. Then, a general framework is proposed for quantification of random and interval uncertainties in the mechanism. In the CRMTM, a transformation method is applied to transform intervals into deterministic arrays,

**Table 4 Uncertain parameters of the deployable mechanism for different conditions**

| Parameter | Case 1 | Case 2 | Case 3 |
|---|---|---|---|
| Clearance size (mm) | {0.2, 0.35, 0.5} | $U$(0.2, 0.5) | {0.2, 0.35, 0.5} |
| Length of link $OA_i$ (mm) | $U$(399.8, 400.2) | $U$(399.8, 400.2) | {399.8, 400, 400.2} |
| Length of link $A_jP_j$ (mm) | $U$(1599.8, 1600.2) | $U$(1599.8, 1600.2) | {1599.8, 1600, 1600.2} |



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while probability theory is used to obtain the 95% confidence region, under the condition of given fuzzy values. The confidence region, considering random and fuzzy uncertainties, is obtained by fuzzy set theory. The numerical examples lead to the following observations. The accuracy of the CRMQRFU is satisfied for an appropriate number used for sampling of the fuzzy uncertainty. The accuracy of the CRMTM is also acceptable. The reliability of the results obtained by the CRMQRFU and CRMTM are verified. Comparing with fuzzy set theory, the proposed methods can improve prediction accuracy of the dynamic performance of the mechanisms, while the results obtained by the proposed methods are more reliable than that produced by the probabilistic approach. The membership function of the clearance size can obviously affect the results. In conclusion, two general methods are presented for accurately predicting the dynamic responses of mechanisms with fuzzy joint clearance and random geometry.

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## Nomenclature

$\tilde{A}$ = fuzzy set
$\mathbf{a}_N$ = random variables
$\mathbf{a}_i$ = random sample
$a_i^{(j)}$ = lower bound of the interval at the membership level $\mu^j$ for the $i$th fuzzy number
$b_i^{(j)}$ = upper bound of the interval at the membership level $\mu^j$ for the $i$th fuzzy number
$c$ = radial clearance size
$c_d$ = dynamic correction coefficient
$c_e$ = restitution coefficient
$c_f$ = friction coefficient
$D_{\text{mod}}$ = modified damping coefficient
$e_{ij}$ = magnitude of eccentricity vector
$\mathbf{e}_{ij}$ = eccentricity vector
$E_k(k=i,j)$ = Young's modulus
$\mathbf{e}_M$ = fuzzy numbers
$\mathbf{e}_M^{\alpha}$ = $\alpha$-cut for fuzzy numbers
$\mathbf{F}_{\text{C}}$ = contact force
$F_{\text{N}}$ = normal contact force
$F_{\text{T}}$ = tangential friction force
$K_n$ = nonlinear stiffness coefficient
$l_1$ = crank length
$l_2$ = connecting rod length
$\mathbf{M}$ = generalized mass matrix
$\mathbf{n}$ = unit eccentricity vector
$\mathbf{q}$ = global generalized coordinates
$\ddot{\mathbf{q}}$ = generalized acceleration vector
$\mathbf{Q}^{\text{A}}$ = external force
$R_i$ = radial of the bearing
$R_j$ = radial of the journal
$\mathbf{r}_k^p(k=i,j)$ = position vectors of $P_k(k=i,j)$ defined in the global coordinates
$\mathbf{r}_k^Q(k=i,j)$ = position of contact point $Q_k(k=i,j)$
$\dot{\mathbf{r}}_k^Q(k=i,j)$ = velocity of contact point $Q_k(k=i,j)$
$v_k(k=i,j)$ = Poisson's coefficient
$v_{\text{N}}$ = relative scalar normal velocity
$\mathbf{v}_T$ = relative tangential velocity
$v_{\text{T}}$ = relative scalar tangential velocity
$\bar{\alpha}$ , $\beta$ = Baumgarte parameter
$\boldsymbol{\lambda}$ = Lagrange multiplier vector
$\delta$ = relative penetration depth
$\dot{\delta}$ = relative penetration velocity
$\dot{\delta}^{(-)}$ = initial penetration velocity
$\boldsymbol{\Phi}_{\mathbf{q}}$ = constraint Jacobian matrix
$\mu_{\tilde{A}}$ = membership level
$\omega$ = angular velocity

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